C10.2 Master requirements to support snoop filters

The snoop filter monitors the snoop address and the snoop response channels to determine the effect of snoop transactions on the allocation of particular cache lines, and the IsShared response is used to determine if a cache line remains allocated in a cache after a snoop. A master component that is implementing snoop filter functionality must therefore provide an accurate IsShared response. See *Read response signaling* on page C3-167.

A master component must ensure that the transactions issued never indicate to the external snoop filter that a cache line is not allocated when the master still holds a copy of the cache line. A master component can ensure that the snoop filter always has correct allocation information using the following techniques.

This specification recommends a master component must not:

- issue a transaction that indicates that the cache line is to become allocated, while a transaction that indicates the same cache line is to be evicted is in progress
- issue a transaction that indicates that a cache line is to be evicted, while a transaction that indicates the same cache line is to be allocated is in progress.

If a master component does overlap allocating and evicting transactions then the following must apply:

- from the first cycle that there is an overlapping allocating transaction and evicting transaction for the same cache line, the cached master must not have the line allocated
- the line must remain de-allocated until a non-overlapping allocating transaction has completed
- when the overlapping allocating and evicting transactions have all completed the allocation status of the line must be resolved, by issuing either:
  - an allocating transaction, to indicate the line is allocated
  - an evicting transaction, to indicate the line is de-allocated.

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